

SMURFS POLYTECHNIC UNION

GCL mid e

Which of the following will dissolve in water (H_2O) to produce a clear solution

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A. $CaCl_2$ ☐

B. $AgBr$ ☐

C. KBr ☒

D. $CH_3(CH_2)_8OH$ ☐

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Quiz GC

To determine the limiting reactant in a chemical reaction, one must know the

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- A. speed of the reaction ☐
- B. available amount of each of each reactant ☐
- C. available amount of one of the reactants. ☐
- D. amount of product formed ☐

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Based on the molecular view of the chemical reaction between 6 moles of H_2 and 4 moles of Cl_2 , which is the excess reactant



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A. HCl ☐

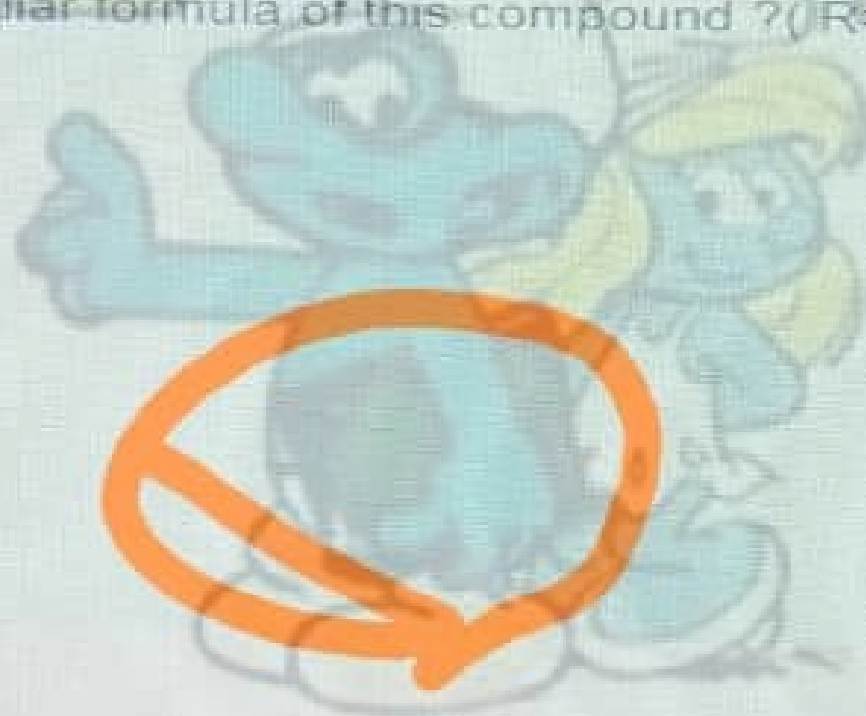
B. H_2 ☐

C. They are both limiting reactants ☐

D. Cl_2 ☐

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A compound has the empirical formula of SF_4 (108 g/mol). At 20°C , 0.300 g of the gaseous compound occupies a volume of 22.1 ml with 1.02 atm. What is the molecular formula of this compound? ($R=0.08206\text{ L atm/K.mol}$)



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- a. SF_4 ☐
- b. S_2F_8 ☐
- c. S_4F_{16} ☐
- d. S_3F_{12} ☒

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A 9.294 g sample of a compound was analyzed for nitrogen N ($N=14\text{g/mol}$). In the procedure, all the nitrogen present was completely converted to ammonia ($\text{NH}_3=17\text{g/mol}$, $\text{H}=1\text{g/mol}$). 3.196 grams of ammonia were obtained. The percent, by mass, of nitrogen in the compound is therefore

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a. 36.0% ☐

b. 28.3% ☐

c. 61.2% ☒

d. 49.7% ☐

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In which experiment dose this apparatus used ?



Select one:

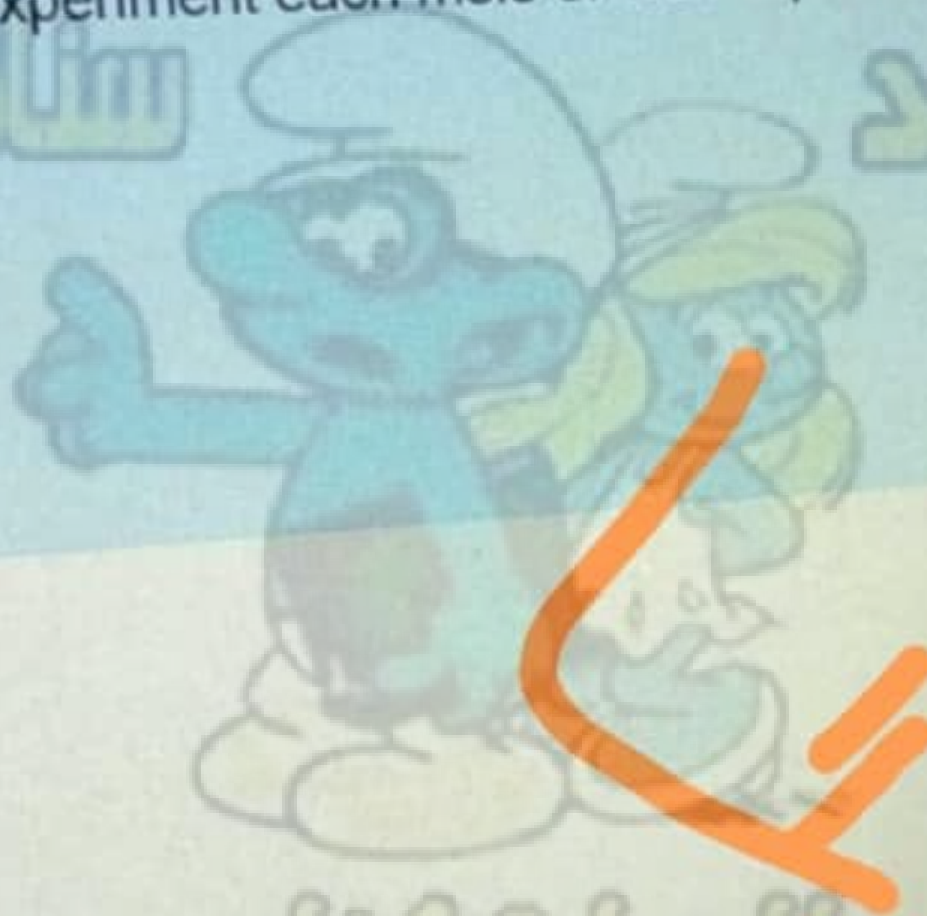
- ☐ A. Empirical formula of ionic compound
- ☐ B. Boiling point identification
- ☐ C. Water of crystallization
- ☐ D. Molecular weight of volatile liquid

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In bleach titration experiment each mole of OCl^- equal one mol of $\text{S}_2\text{O}_3^{2-}$

Select one:

- ☐ True
- ☐ False



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The % W/W of NaOCl in a bleach sample equal 6%. The mass of NaOCl in 2L sample of this bleach ($d=1.084 \text{ g/mL}$) equals

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- a. 130.1g ☒
- b. 151.76g ☐
- c. 109.3g ☐
- d. 173.4g ☐

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For the reaction below, the suitable rate law expression is:



Select one:

- ☐ a. $R = k [\text{NO}]^x [\text{CO}_2]^y$
- ☐ b. $R = k [\text{NO}]^x [\text{NO}_2]^y$
- ☐ c. $R = k [\text{CO}]^x [\text{CO}_2]^y$
- ☐ d. $R = k [\text{NO}_2]^x [\text{CO}]^y$



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one of the unit of Enthalpy of reaction is (K.mol)

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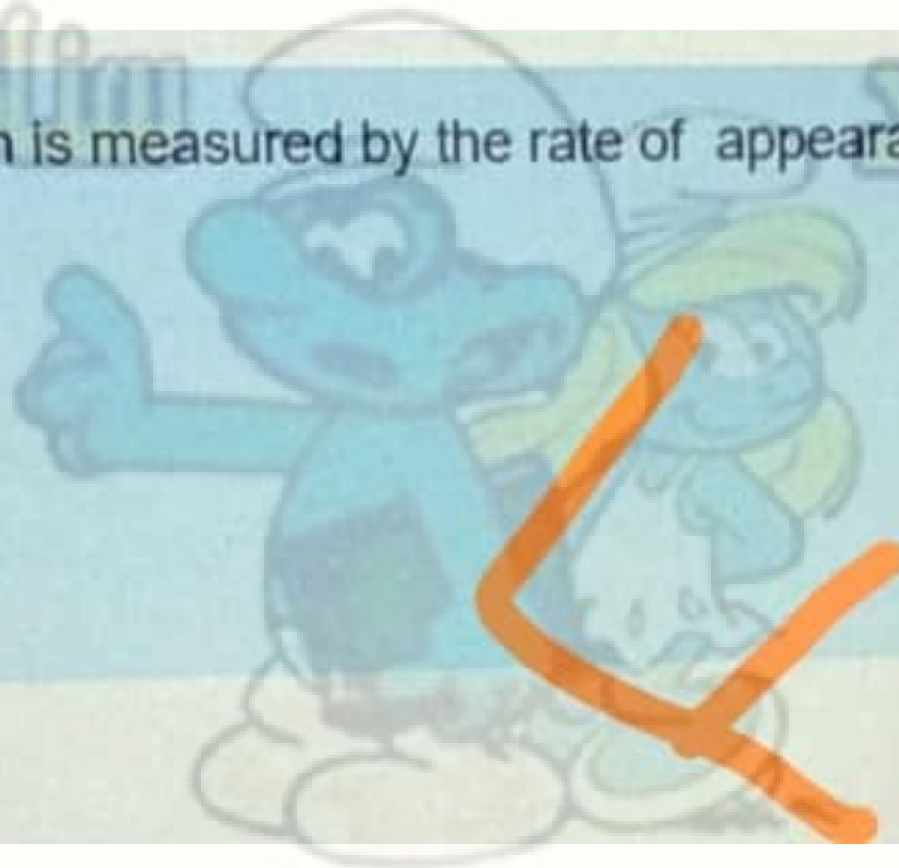
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The rate of reaction is measured by the rate of appearance of product with time

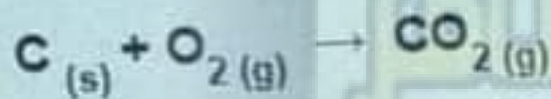
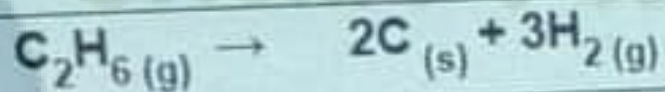
Select one:

- ☐ True
- ☐ False

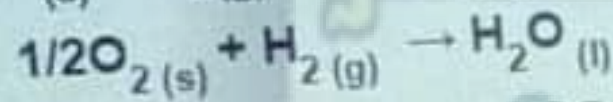


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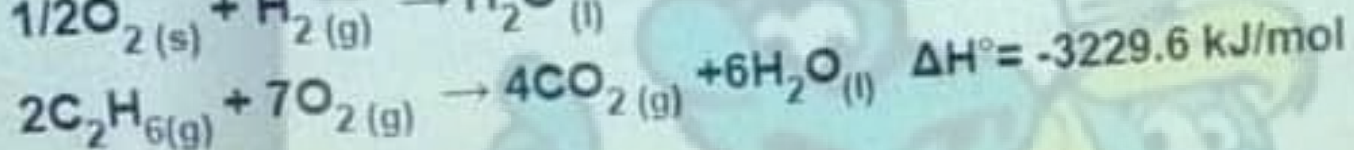
Use the following data to calculate the enthalpy change, in (kJ mol^{-1}), for the reaction in the box.



$$\Delta H^\circ = -293.5 \text{ kJ/mol}$$



$$\Delta H^\circ = -285.5 \text{ kJ/mol}$$



$$\Delta H^\circ = -3229.6 \text{ kJ/mol}$$

Select one:

- ☐ a. 28.7
- ☐ b. 171.3
- ☐ c. -171.3
- ☐ d. -28.7

The following neutralization reaction occurs in the classroom.



If a student uses 25.0 mL of a 0.5M solution of KOH, what is the molarity of the acid if 15.0mL of acid neutralized?

Select one:

- ☐ A. 12.5M
- ☐ B. 1.2M
- ☐ C. 1.25 M
- ☐ D. 0.83M

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During experiment of chemical kinetics for the reaction $\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$, the rate law expression is $\text{Rate} = k [\text{H}_2\text{O}_2]^y [\text{I}^-]^x$. Use the data in table to calculate x

Run#	$[\text{I}^-] \text{ M}$	$[\text{H}_2\text{O}_2] \text{ M}$	Rate (M/s)
1	0.22	0.064	0.24
2	0.88	0.064	0.96
3	0.22	0.032	0.24

Select one:

- ☐ a. $x=2$
- ☐ b. $x=1$
- ☐ c. $x=0$
- ☐ d. $x=3$

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Consider the set up on the right which used during vinegar analysis. The names of equipment 1 and 2 are



Select one:

- ☐ a. 1-flask
2-test tube
- ☐ b. 1-Burette
2-flask
- ☐ c. 1-Flask
2-Burette
- ☐ d. 1-beaker
2-test tube

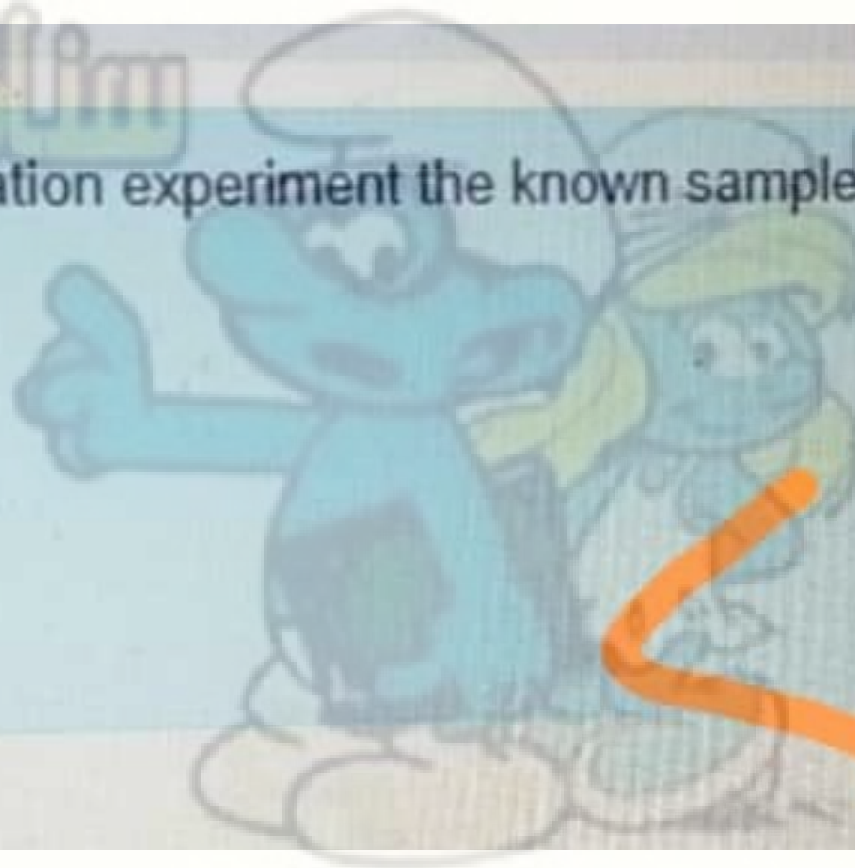
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In acid- base titration experiment the known sample is put in flask

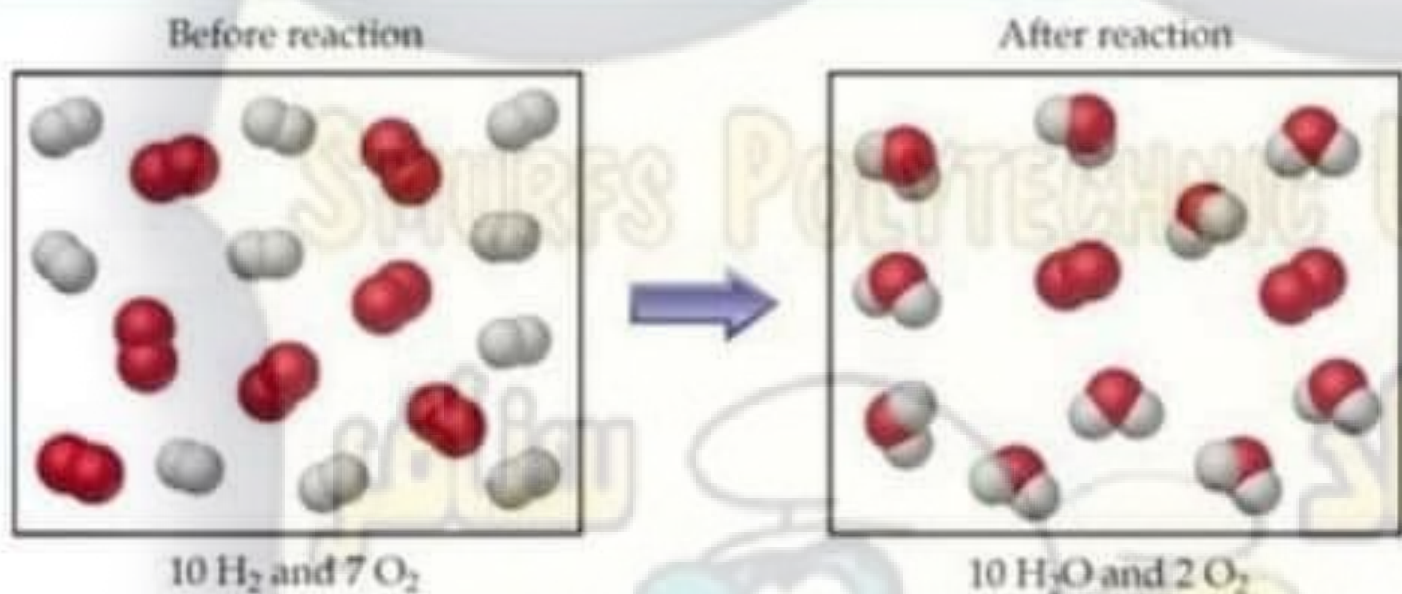
Select one:

- ☐ True
- ☐ False



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Based on the molecular view of the chemical reaction between 10 moles of H_2 and 7 moles of O_2 , which is the limiting?



Select one:

- ☐ A. H_2O
- ☐ B. both are excesses
- ☐ C. H_2
- ☐ D. O_2

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In acid- base titration experiment the color of acidic medium is pink

Select one:

- ☐ True
- ☐ False

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In bleach analysis experiment, 1.0 g of KI was added to the diluted bleach sample to reduce:

Select one:

- ☐ A. H_2O
- ☐ B. Na^+
- ☐ C. H^+
- ☐ D. OCl^-

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Hypochlorite, ClO^- (aq), is the bleaching agent in bleach.

Select one:

- ☐ True
- ☐ False

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In Endothermic reaction, solution gained heat from the reaction.

Select one:

- ☐ True
- ☐ False

ΔH of reaction equal ($-\Delta H$) of solution

Select one:

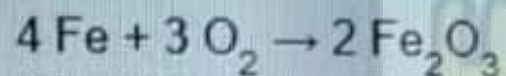
- ☐ True
- ☐ False



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If 2.1453 moles of Fe are reacted with 4.4434 moles of O_2 , how many moles of Fe_2O_3 will be produced?

Select one:

- ☐ A. 1.4811 mol
- ☐ B. 0.5363 mol
- ☐ C. 2.9622 mol
- ☐ D. 1.0727 mol

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The % W/W of NaOCl in a bleach sample equal 7%. The mass of NaOCl in 2L sample of this bleach ($d=1.084 \text{ g/mL}$) equals:

Select one:

- ☐ a. 151.76g
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- ☐ c. 130.1g
- ☐ d. 109.3g

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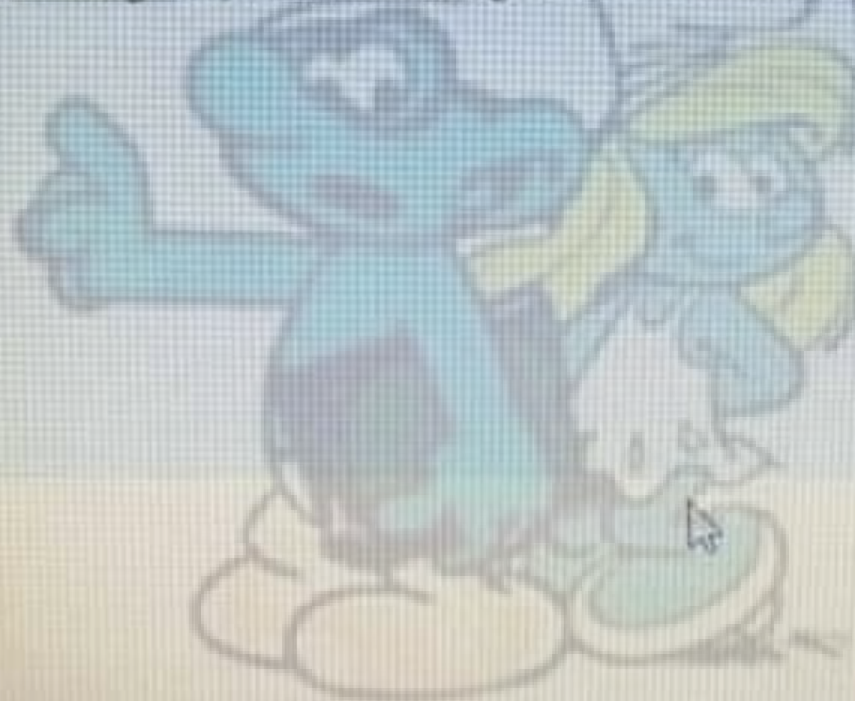
Ammonia burns in the presence of Platinum catalyst to Nitric oxide :



In an experiment, 68 g of NH_3 (Mm= 17 g/mol) is burned and give 11.7 KJ/mol of enthalpy. the value of heat (q_{rxn}) in joule is:

Select one

- ☐ A. $+4.68 \times 10^4 \text{ J}$
- ☐ B. $-4.68 \times 10^3 \text{ J}$
- ☐ C. $+4.68 \times 10^3 \text{ J}$
- ☐ D. $-4.68 \times 10^4 \text{ J}$



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What is the greatest amount of MgO (in moles) that can be made of 7.8 moles Mg and 3.7 moles of O?



MgO

Select one:

- ☐ a. 7.8 moles
- ☐ b. 3.7 moles
- ☐ c. 7.4 moles
- ☐ d. 3.9 moles

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for the reaction: $P_4 + 6Cl_2 \rightarrow 4PCl_3$ if 75.0g P_4 react with excess chlorine gas produces 210g PCl_3 in lab. Calculate percent yield for the reaction. ((if the theoretical yield = 333 g)



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A. 23% ☐

B. 85% ☒

C. 63% ☐

D. 36% ☐

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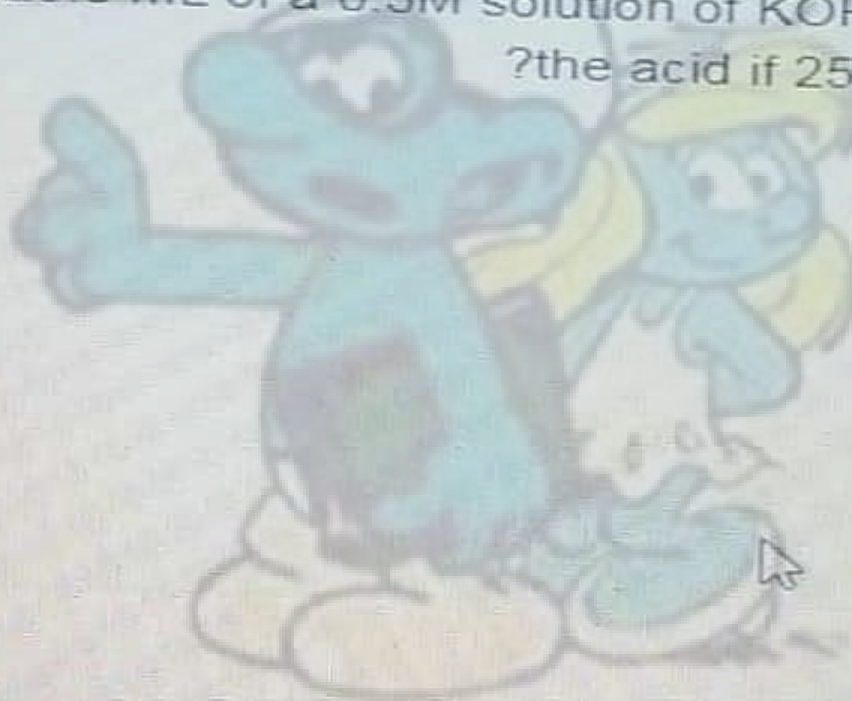
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Quiz GCL

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If a student uses 25.0 mL of a 0.5M solution of KOH, what is the molarity of the acid if 25.0mL of acid neutralized



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- A. 0.5 M ☒
- B. 0.8M ☐
- C. 12.5M ☐
- D. 1.2M ☐

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